

Assignment - 2

Lists Operations

Do the following in sequence and record the results in a single program Create a list with 5 different types of fruits. Print the list. Add two more fruits to the list, then remove one fruit from it. Print the updated list. Access the second and fourth fruits in the list. Print them. Slice the list to get the first three fruits and print the result. Find and print the length of your list.

```
#List Operations

fruits = ['apple','banana','orange','pineapple','mango']
print('The list of fruits are: ',fruits)

fruits.extend(['watermelon','cherry'])
print('The list after adding two fruits: ',fruits)

fruits.remove('banana')
print('The list after removing one fruit: ',fruits)

second_fruit = fruits[1]
fourth_fruit = fruits[3]
print('The second fruit in the list is: ',second_fruit)
print('The fourth fruit in the list is: ',fourth_fruit)

sub_fruits = fruits[0:3]
print('The first three fruits are: ',sub_fruits)

length = len(fruits)
print('The length of the list is: ',length)
```

Output:

```
"C:\Wipro Training\Python\PythonBasics\venv\Scripts\python.exe" "C:\Wipro
Training\Python\PythonBasics\Assignments\day_1\lists.py"
The list of fruits are: ['apple', 'banana', 'orange', 'pineapple', 'mango']
The list after adding two fruits: ['apple', 'banana', 'orange', 'pineapple', 'mango', 'watermelon',
'cherry']
The list after removing one fruit: ['apple', 'orange', 'pineapple', 'mango', 'watermelon', 'cherry']
The second fruit in the list is: orange
The fourth fruit in the list is: mango
The first three fruits are: ['apple', 'orange', 'pineapple']
The length of the list is: 6
```

Tuple Operations

Do the following in sequence and record the results in a single program. Create a tuple with the names of 3 different cities you'd like to visit. Print the tuple.: Access and print the first and last elements of the tuple.: Create another tuple with 2 more cities. Concatenate both tuples and print the result. Try changing one element of the tuple. What happens? Unpack the elements of the tuple into separate variables and print them.

```
#tuple operations

cities = ('Tokyo','Paris','Seoul')
print('The cities i would like to visit are: ',cities)

first_city = cities[0]
last_city = cities[-1]
print('The first city in my list is: ',first_city)
print('The last city in my list is: ',last_city)

cities2 = ('London','Bangkok')
concatated_cities = cities + cities2
print('The Concatenation of two tuples is: ',concatated_cities)

cities[0] = 'Kyoto'
print('The list after changing the first city to Kyoto: ',cities[0])

(city1, city2, city3, city4, city5) = concatated_cities
print('For variable "city1" the value is: ',city1)
print('For variable "city2" the value is: ',city2)
print('For variable "city3" the value is: ',city3)
print('For variable "city4" the value is: ',city4)
print('For variable city5 he value is: ',city5)
```

Output:

"C:\Wipro Training\Python\PythonBasics\venv\Scripts\python.exe" "C:\Wipro Training\Python\PythonBasics\Assignments\day_1\tuples.py"

The cities i would like to visit are: ('Tokyo', 'Paris', 'Seoul')

The first city in my list is: Tokyo

The last city in my list is: Seoul

The Concatenation of two tuples is: ('Tokyo', 'Paris', 'Seoul', 'London', 'Bangkok')

Traceback (most recent call last):

File "C:\Wipro Training\Python\PythonBasics\Assignments\day_1\tuples.py", line 15, in <module>

```
cities[0] = 'Kyoto'
```

```
~~~~~^
```

TypeError: 'tuple' object does not support item assignment

Process finished with exit code 1

```
#tuple operations

cities = ('Tokyo','Paris','Seoul')
print('The cities i would like to visit are: ',cities)

first_city = cities[0]
last_city = cities[-1]
print('The first city in my list is: ',first_city)
print('The last city in my list is: ',last_city)

cities2 = ('London','Bangkok')
concatated_cities = cities + cities2
print('The Concatenation of two tuples is: ',concatated_cities)

# cities[0] = 'Kyoto'
# print('The list after changing the first city to Kyoto: ',cities[0])

(city1, city2, city3, city4, city5) = concatated_cities
print('For variable "city1" the value is: ',city1)
print('For variable "city2" the value is: ',city2)
print('For variable "city3" the value is: ',city3)
print('For variable "city4" the value is: ',city4)
print('For variable "city5" the value is: ',city5)
```

Output - 2(Without assignment)

"C:\Wipro Training\Python\PythonBasics\venv\Scripts\python.exe" "C:\Wipro Training\Python\PythonBasics\Assignments\day_1\tuples.py"

The cities i would like to visit are: ('Tokyo', 'Paris', 'Seoul')

The first city in my list is: Tokyo

The last city in my list is: Seoul

The Concatenation of two tuples is: ('Tokyo', 'Paris', 'Seoul', 'London', 'Bangkok')

For variable "city1" the value is: Tokyo

For variable "city2" the value is: Paris

For variable "city3" the value is: Seoul

For variable "city4" the value is: London

For variable "city5" the value is: Bangkok

Set Operations

Do the following in sequence and record the results in a single program. Create a set of 5 unique colors. Print the set. Add a new color to the set, then try removing an existing color. Print the updated set. Create another set with 3 different colors. Find the intersection, union, and difference between both sets. Check if a specific color is in the set and print the result. Convert a list of fruits (with some duplicates) into a set and print the unique fruits.

```
#Set Operations

colours = {'Red', 'Blue', 'Green', 'Yellow', 'Purple'}
print('My set of colours are: ', colours)

colours.add('Orange')
colours.remove('Yellow')
print('The set after adding new colour and remove another colour: ', colours)

new_colours = {'Black', 'Maroon', 'Peach'}
print('The intersection of two sets is: ', colours.intersection(new_colours))
print('The union of two sets is: ', colours.union(new_colours))
print('The difference between two sets are: ', colours.difference(new_colours))

if 'Red' in colours:
    print('Yes!! Red is in the set')
else:
    print('No!! Red is not in the set')

fruits = ['orange', 'mango', 'cherry', 'orange', 'apple']
print('The list of fruits are: ', fruits)

new_fruits = set(fruits)
print('The set of fruits are: ', new_fruits)
```

Output:

```
"C:\Wipro Training\Python\PythonBasics\venv\Scripts\python.exe" "C:\Wipro Training\Python\PythonBasics\Assignments\day_1\sets.py"
```

```
My set of colours are: {'Green', 'Yellow', 'Blue', 'Red', 'Purple'}
```

```
The set after adding new colour and remove another colour: {'Green', 'Orange', 'Blue', 'Red', 'Purple'}
```

```
The intersection of two sets is: set()
```

```
The union of two sets is: {'Green', 'Orange', 'Maroon', 'Blue', 'Black', 'Peach', 'Red', 'Purple'}
```

The difference between two sets are: {'Green', 'Orange', 'Blue', 'Red', 'Purple'}
Yes!! Red is in the set
The list of fruits are: ['orange', 'mango', 'cherry', 'orange', 'apple']
The set of fruits are: {'apple', 'cherry', 'orange', 'mango'}

Process finished with exit code 0

Dictionary Operations

Do the following in sequence and record the results in a single program. Create a dictionary with your name, age, and favorite hobby as keys, and provide appropriate values. Print the dictionary. Access and print the value associated with your name in the dictionary. Add a new key-value pair for your favorite food, then update the value for your favorite hobby. Print the updated dictionary. Print all the keys and all the values in the dictionary separately. Remove the age key-value pair from the dictionary. Print the dictionary to see the change.

```
#Dictionary Operations

my_details={'Name': 'Durgam Deekshita',
            'Age': 23,
            'Hobby': 'Reading Books'}
print('My details are: ',my_details)

name = my_details['Name']
print('My name is: ',name)

my_details['Favourite_food'] = 'Biryani'

my_details['Hobby'] = 'Vibe coding'
print('The updated details are: ',my_details)

print('All the keys are: ',list(my_details.keys()))
print('The values of all the keys are: ',list(my_details.values()))

age = my_details.pop('Age')
print('The updated details after removing age key: ',my_details)
```

Output:

"C:\Wipro Training\Python\PythonBasics\venv\Scripts\python.exe" "C:\Wipro Training\Python\PythonBasics\Assignments\day_1\dictonaries.py"

My details are: {'Name': 'Durgam Deekshita', 'Age': 23, 'Hobby': 'Reading Books'}

My name is: Durgam Deekshita

The updated details are: {'Name': 'Durgam Deekshita', 'Age': 23, 'Hobby': 'Vibe coding', 'Favourite_food': 'Biryani'}

All the keys are: ['Name', 'Age', 'Hobby', 'Favourite_food']

The values of all the keys are: ['Durgam Deekshita', 23, 'Vibe coding', 'Biryani']

The updated details after removing age key: {'Name': 'Durgam Deekshita', 'Hobby': 'Vibe coding', 'Favourite_food': 'Biryani'}

Process finished with exit code 0